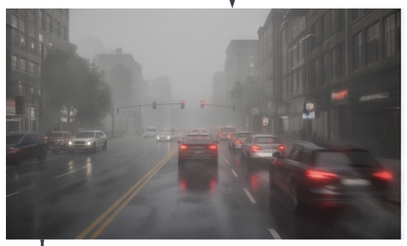
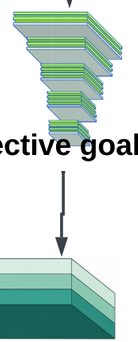
1: Rollout

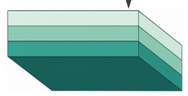
Input from CARLA-SUMO



Dense Smog & Heavy Rain



Multi-objective goal sampler

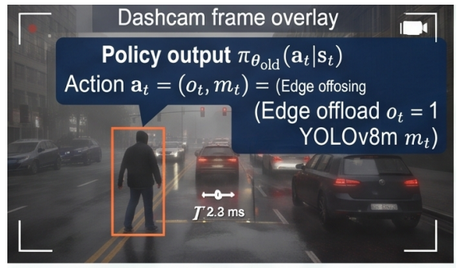


Choosing goal samle

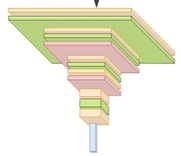
Inputs

- Previous frame's dual-head XAI feedback
- RSU Queue delay
- State
- Link rate

2: Policy Action



Dashcam frame overlay
- Policy output
- Action (Edge offload , YOLOv8m)
- The offloading transmission (and edge execution



High-gated uncertainty

Output Data

Dual-head XAI



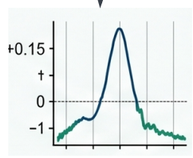
- Accuracy
- Drop indicator
- Violation

Scalar Reward orn multi-objective loss

3: Optimizaton Pipeline

Advantage & Value Check

- Advantage
- Value error



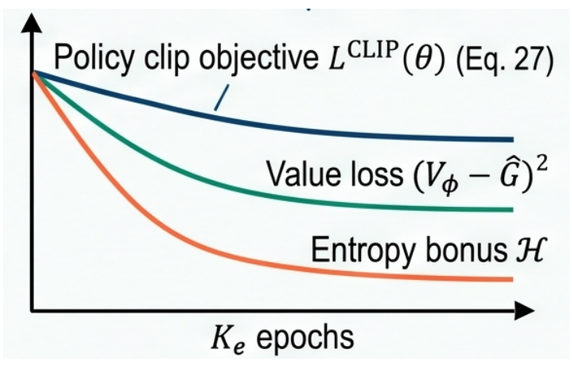
- Advantage normalisation graph

Loss Function Math / Graph

Gradien

Mini-batches

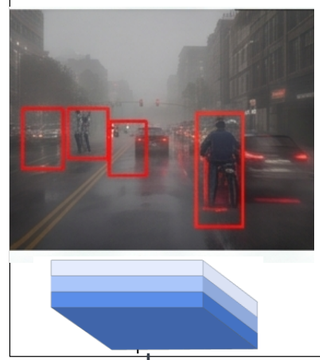
rollout gradients



4: Training Improvement On Validation Frame

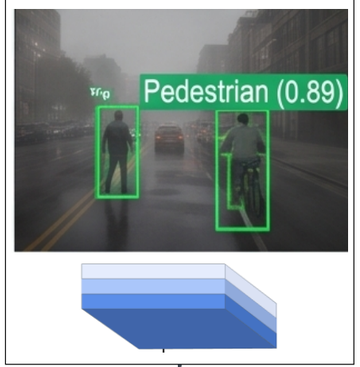
Side wiving Outcome of the gradient update on a single validation test image

Old Policy



- Incorrect/Loose detections
- Latency over threshold
- High validation loss

Updated Policy



- Highly focused activations
- Precise labeling
- (No ghosting, no delay)

- E2E Task Latency
- Deadline violations
- Workload Energy
- Delivered Accuracy